

Energy efficiency as a sustainability concern in Africa and financial development: How much bias is involved?

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ABSTRACT

This study contributes to the literature on whether financial development stimulates technical energy efficiency (TEE) or not, by addressing core biases that creep into the relationship and thereby reducing the ability to draw causal inferences from financial development to TEE. Our approach is based on the instrumental stochastic frontier technique, where biases in the frontier and inefficiency equations are dealt with using external instrumental variables. The legal system origin of the country and life expectancy at birth were used as instruments for financial development and income, respectively. The current study demonstrates substantial bias in income elasticity, the estimate of energy efficiency, and the effect of financial development on energy efficiency. Both income elasticity and energy efficiency estimate risk upward bias. Equally, the effect of financial system development and financial institution development on TEE risk downward bias. Other results show that all aspects of financial institution development stimulate TEE, but access to financial institutions is more important. These results raise caution about future studies' estimates of the effect of financial development on TEE. Though this study has demonstrated the potency of external instruments in dealing with the bias in the coefficient estimates, we consider this might prove to be a luxury solution in some cases due to data limitation, context differences, and theory. In those circumstances, reliance on internal instruments might prove to be the second-best option.

1. Introduction

Globally, energy intensity improved by 2% from 2018 to 2019. However, energy savings due to technical improvement reduced by 5% in 2019 relative to 2018 (International Energy Agency, [IEA], 2020). Countries in Africa continue to struggle with high energy inefficiency due to poor institutions, lack of regulation, and undeveloped markets among others (Adom and Adams, 2018; Adom, 2016; Adom et al., 2018). This situation is very worrying given that energy efficiency improvement alone could be responsible for about 40% reduction in energy-related emissions (IEA, 2020). Particularly, in Africa, where energy security is a major concern, delivering energy efficiency goals is key to the continent's attempt to address energy poverty issues. As noted by the International Energy Agency, investing US\$1 in demand-side

energy efficiency can save more than US\$2 on the supply side (IEA, 2020).

Due to the hefty nature of the investment required in energy-efficient technologies, the availability of secure funds is key to achieving the energy efficiency goal. However, while countries in Europe have announced to commit about US\$15 billion to support energy efficiency-related investments, in Africa, governments are yet to announce a budget commitment to support energy efficiency-related investments (IEA, 2020). The situation is worsened by the fact that, in Africa, access to credit is problematic, and even when possible, it is done at a very high cost. These conditions generally inhibit investment in technology. Developing the financial sector is key to achieving a low cost of capital and enhancing access to credit, which are necessary conditions to boost investment (Sviryzdenka, 2016). As noted by Benfratello et al. (2008)

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and Amore et al. (2013), the growth of the financial sector moves parallel with technological innovation. Bearing this in mind, this study investigates the role of different aspects of financial system development in influencing technical energy efficiency (TEE) in Africa based on a stochastic frontier technique.

The existing literature on energy and finance ignores key identification issues. This introduces bias in the coefficient estimates and limits the ability of the models to infer a causal link from financial development to energy outcomes. This study contributes to this literature by exploring how key identification issues contribute to bias in the estimate of income elasticity in the frontier equation, the estimate of TEE, and financial development effect in the inefficiency equation. The first bias relates to measuring real gross domestic product per capita in the frontier equation. National income accounts data in Africa have been reported to be of poor quality (Adom and Amoani, 2021). Official gross domestic product may also be underreported due to the large informal sector. This implies that unobserved factors that correlate with income and explain variation in the outcome variable (i.e. energy consumption) will introduce bias in the coefficient estimate of income. Also, economic growth may not be decoupled from the environment and hence energy resources, and this interdependence could cause reverse causality and introduce bias in the income estimate. The introduction of bias in the frontier equation implies that the frontier may be incorrectly estimated, and this could introduce bias in the estimate of TEE.

The second bias originates from measuring financial development. A large number of empirical studies use narrow-based measures of financial development (see: Adom, 2021b; Liao and Drakeford, 2019; Chen et al., 2019b; Chen et al., 2019a; Amuakwa-Mensah et al., 2018, inter alia). The high correlation existing between different dimensions of financial development makes these narrow-based measures biased negatively or positively. Because, it is the constellation of financial markets and financial institutions that promote the provision of financial services (Sviryzdenka, 2016), broad-based measures of financial development are preferred to narrow-based measures (see Sviryzdenka, 2016). Comparatively, the extent of bias is limited with broad-based measures of financial development. However, the use of broad-based measures of financial development (see: Canh et al., 2020; Bashir et al., 2020) does not eliminate completely the bias inherent in the measurement of financial development. Sviryzdenka's (2016) broad-based financial development index, which is used in this study, for example, does not capture all aspects of financial sector operations such as credit transfer and mobile money banking. Further, the index ignores the diverse and complex nature of financial intermediation and financial institution. Lastly, the index depicts more of the characteristics of the financial sector instead of outcomes of the financial sector. These factors, if correlated with variations in the outcome variable (TEE), would introduce bias in the coefficient estimate of financial development in the inefficiency equation.

In addition to the above biases, there are two more sources of bias that have not been adequately addressed in the literature. The first is due to simultaneity bias. Improvements in TEE trigger a reduction in energy bills (Duan et al., 2017). Three possibilities can emerge. First, if the energy bills saved are channelled to financial savings, it can increase the supply of loanable funds and hence improve the credit business of the financial sector, all things being equal. Second, alternatively, the energy bills saved by the energy user can be used to repay outstanding loans and help improve non-performing loans in the financial sector. Adom (2019) and Adom et al. (2020a) have shown that increasing energy consumption efficiency triggers financial sector development and bank profit, respectively. Sleuwagenn and Goedhuys (2003) also note that efficiency gains in inputs can make banks become more competitive and profitable and increase their market share. Consequently, higher energy efficiency levels are expected to increase financial development. Therefore, in a regression of energy efficiency and financial development, the coefficient of financial development will suffer from upward bias. Third, savings made from energy bills due to improvement in TEE can either

intensify the use of existing energy-using appliances or stimulate investment in new energy-using appliances. If these rebound effects associated with the former case are great, increases in TEE would imply a loss of loanable funds and lower development of financial sector. In this case, regressing TEE on financial development will produce a downward bias in the coefficient of financial development. In the latter case, investment in new energy-efficient technologies can enhance the degree of financialisation of the financial sector. Therefore, in regressing energy efficiency on financial development, the coefficient of financial development may suffer from an upward bias.

The last source of bias relates to the assumption of TEE. Largely, studies on TEE have focused on short-term (i.e. annual, quarterly, or monthly) variations and the possible underlying causes (Liu et al., 2023; Shen and Lin, 2017; Edziah et al., 2021; Sun et al., 2021, inter alia). This assumes that TEE exhibits rapid changes in the short-term. However, for some countries, at least, TEE is not expected to exhibit rapid changes in the short-term. There are a number of studies that have emphasised the persistent nature of energy efficiency/inefficiency (Filippini and Zhang, 2016; Alberini and Filippini, 2018; Adom et al., 2018; Adom and Adams, 2018; Adom et al., 2021; Agradi et al., 2022, inter alia). Indeed studies by Adom (2016), Adom and Adams (2018), and Zhang and Adom (2018) emphasised that for developing economies like Africa and China, the energy efficiency transition tendency is likely in the long-run. This is because a switchover from one technology to another may be practically difficult. Also, different energy users may use varying discount rates and face different market imperfections and barriers. Modelling the relationship between energy efficiency and financial development should take into account the possibility that energy efficiency does not exhibit very rapid changes in the short-term. Failing to do so will introduce bias and affect the causal inference from financial development to TEE.

This study contributes to the literature by addressing the biases identified. Our results are based on the endogenous stochastic frontier approach, where income (in the frontier equation) and financial development (in the inefficiency equation) are endogenous. We deal with the bias associated with measuring income and financial development by relying on the instrumental variable technique. Life expectancy at birth is used to instrument real gross domestic product per capita in the frontier equation while the legal system origin of the country is used to instrument financial development. Also, we perform regressions at a much higher order to address concerns that TEE may not exhibit very rapid changes in the short-term. Given these controls, our results raise caution about previous studies' estimate of the effect of financial development on TEE. Our results have demonstrated that significant bias exists in the estimate of income elasticity, TEE, and the effect of financial development on TEE. We find substantial upward bias in the estimate of income elasticity and TEE. The result further shows that financial system development and financial institution development stimulate TEE but these estimates risk downward bias. All aspects of financial institution development stimulate TEE, but more importantly, is access to financial institutions. These results reiterate the need for future studies on energy and finance to consider identification issues critical. While the current study proposes the use of external instruments, it might be considered as luxury in some cases. Therefore, we caution that there is no one-size-fits-all solution, as context and availability of appropriate instruments may be critical in the decision-making.

The rest of this article is organised as follows: Section 2 discusses the literature. Section 3 describes the method and data, section 4 presents the results, and section 5 concludes with policy implications.

2. Literature review

Two theories of change - detailing the channels by which the effects of financial development transmit to TEE - have been proposed in the empirical literature. The first theory of change (i.e. scale effect) is derived from the theoretical link existing between economic

development and financial development (Beck et al., 2004; Beck et al., 2000; Calderón and Liu, 2003). The scale effect asserts that financial sector development aids the financing of energy-consuming appliances and the creation of new products, which might be energy-intensive (Sadorsky, 2010). These scale impacts, all things being equal, are expected to increase energy consumption and hence decrease energy consumption efficiency. The scale effect assumes that the technological status quo does not change with the scale expansion. This is disputed by the second theory of change, which is the technical effect. It asserts that the growth of the financial sector stimulates investment in technological innovation, changing the existing technology. Technological innovation improves the technical efficiency of inputs, which includes energy resources (Shahbaz and Lean, 2012; Adom, 2021a).

The empirical literature is also not conclusive. Bashir et al. (2020) find that financial system development improves energy efficiency for OECD countries. Similarly, Adom et al. (2020b) find that financial development improves energy efficiency in Ghana. However, Chen et al. (2019a) find that while financial development improves energy efficiency for non-OECD countries, it exerts a limited impact on energy efficiency for OECD countries. Pan et al. (2019) find that, though financial development improves energy efficiency in Bangladesh, the effect has been reduced lately. Correspondingly, in China, Aller et al. (2018) reveal that financial development improves energy efficiency, but the effect has diminished significantly over time. In the case of Yao et al. (2021) for BRICS and next 11 countries, the authors find that the impact of financial development on energy efficiency is negative but becomes positive when they control for the level of corruption. Amuakwa-Mensah et al. (2018) find that, for Africa, the effect of Commercial Banks development on energy efficiency depends on the institutional environment. Financial development improves energy efficiency in economies where there is centralisation of authority than those with decentralisation of authority. However, Adom (2021a) finds that whether financial development stimulates electricity consumption efficiency in Africa or not depends on the choice of indicator of financial development used.

Central to the energy – finance debate is the choice of indicator for financial development. A large number of studies use narrow-based measures of financial development. Amuakwa-Mensah et al. (2018) use commercial banks' balance sheet performance indicators (i.e. return on asset, asset quality, bank capitalisation, managerial inefficiency, and financial stability). Chen et al. (2019b) and Liao and Drakeford (2019) use bank loans to GDP, market capitalisation to GDP, and depth of insurance to measure financial development. Chen et al. (2019a) use domestic credit to the private sector by banks, private sector credit by banks, and financial openness to proxy financial development. Nasreen and Anwar (2020) use liquid liability as a percent of GDP, domestic credit to the private sector, bank assets as a percent of GDP, and stock market capitalisation to measure financial development. Adom et al. (2020b) use domestic credit to the private sector and broad money supply as a percent of GDP and as a percent of the required reserve ratio as indicators of financial development. Adom (2021a & b) also use depth measures of financial development (i.e. liquid liabilities, financial system deposits, bank assets, and domestic credit to the private sector) in their study on Africa. Yao et al. (2021) use domestic credit to the private sector to proxy financial development. Due to the high internal correlation that might exist among the different indicators of financial development, using more narrowed and specific measures of financial development creates measurement error problems as well as unobserved heterogeneities. Consequently, the estimated effect of financial development on energy efficiency is likely to be biased and inconsistent.

Given the multidimensional aspects of financial development, using broad-based measure seems appropriate. The depth of, access to, and efficiency of financial institutions and markets are important aspects of financial system development that can stimulate energy efficiency improvements. Access deals with the ability of individuals and organisations to access financial services. Depth deals with the size and liquidity

of markets, while efficiency deals with the ability of financial institutions and markets to provide financial services at low cost and with sustainable revenues (Svirydzenka, 2016). There are two studies in the literature on energy and finance that apply Svirydzenka's multidimensional measure of financial development. Canh et al. (2020) analyse the effects of nine disaggregate indicators of financial development (covering access, efficiency, and depth of financial institutions, financial markets, and financial systems) on energy efficiency. Bashir et al. (2020) consider an aggregate indicator of financial development, which is the constellation of financial institutions and markets. Although the Svirydzenka's multidimensional measure of financial development helps address potential measurement errors in financial development, it does not eliminate them completely. This implies that measurement errors creep into both narrow and broad-based measures, but they may be minimised in the latter case. Consequently, the effect of financial development is likely to suffer from bias.

In addition to the measurement concerns, another potential source of bias present in the relationship postulated is simultaneity bias. Improvements in TEE can trigger financial sector development. Previous attempts in the literature (see: Adom et al., 2020b; Nasreen and Anwar, 2020; Amuakwa-Mensah et al., 2018; Aller et al., 2018; Chen et al., 2019a; Canh et al., 2020; Bashir et al., 2020, inter alia) use the lags of financial development to instrument the endogenous financial development variable. However, lag instruments of financial development do not meet the exclusion restriction. Realistically, rational consumers make future consumption based on the current economic and financial conditions. Using external instruments is considered more appropriate when they are available (Adom, 2021b).

Another identification issue is related to measuring energy efficiency. Largely, studies use the energy intensity indicator. Energy intensity is easy to compute and interpret, but variation in energy intensity is driven by social, economic, climatic, and technological factors. This makes it difficult to derive a meaningful interpretation of the effect of financial development on TEE. This is because lower values in energy intensity driven by financial development may not necessarily imply an improvement in energy efficiency. In this regard, a number of studies overcome this problem using either data envelope analytic or stochastic frontier analytic (SFA) methods. However, with the exception of a few studies such as Filippini and Zhang (2016), Zhang and Adom (2018), and Adom (2021b) that adopt pseudo approaches to deal with endogeneity in the frontier equation, the rest of the studies impose exogeneity assumption in the stochastic frontier models. Also, related to TEE, is the assumption that TEE in countries exhibits very rapid changes in the short-term. This might not be true because, for some countries, at least, changes in TEE may not be very rapid, and this could impair causal inference. This study contributes to the literature by demonstrating the extent of bias present in the estimate of TEE, income elasticity, and the effect of financial development. Our approach is based on using an instrumental variable approach and performing regressions at a much higher order.

3. Model specification and data

3.1. Model specification

Theoretically, our estimate of TEE follows the procedure of Kopp (1981). Kopp (1981) uses the non-radial input-specific approach to estimate input-specific technical efficiency as the ratio of the benchmark input use (i.e. minimum feasible input use) to actual input use. Using Kopp's approach, Filippini and Hunt (2015) provide a theoretical economic foundation for TEE. Filippini and Hunt (2015) argue that energy is a derived demand. Therefore, the rational economic agent aims to choose the level of energy input that maximises the provision of energy services. Conceptually, this is similar to a typical production maximisation decision, but, in this case, the economic agent uses energy resources and energy-using appliances as inputs. The empirical

approach used to estimate TEE in this study follows the procedure of Filippini and Hunt (2011, 2012). Filippini and Hunt (2011, 2012) adopt the Kopp non-radial input-specific approach and then proposed an input-specific conditional stochastic energy demand frontier to estimate TEE. The conditional stochastic energy demand frontier (ed_{it}), as shown in eq. (1), consists of the optimal use of energy ($f(X_{it}; \beta)e^{v_{it}}$) and the extent of deviation from the optimal use level ($e^{\mu_{it}}$).

$$ed_{it} = f(X_{it}; \beta)e^{v_{it}} * e^{\mu_{it}} \quad (1a)$$

where X_{it} is a vector of exogenous and endogenous variables in the frontier equation, β is a vector of frontier parameters, v_{it} is the two-sided error term, which is assumed to have a normal distribution, and μ_{it} measures the extent of deviation of actual energy use from the optimal/benchmark level, which is a one-sided error with a half-normal distribution. The one-sided error term.

3.2. Empirical considerations

3.2.1. Identification of frontier equation

Based on standard demand theory, this study includes in the vector X_{it} a measure of energy price (p_{it}), income (gdp_{it}), and temperature ($temp_{it}$). Income captures the effect of economic activities, and temperature reflects the effects of changes in climate. In addition to the listed variables, the study augmented the frontier equation to include time trend (t_{it}) and its square (t_{it}^2) to capture the underlying energy demand trend, which is driven by several other possible effects, such as social and technological innovation (see: Filippini and Hunt, 2011; Filippini and Zhang, 2016; Adom et al., 2018; Zhang and Adom, 2018). Eq. 1b is the log transformation of 1a.

$$\ln ed_{it} = \alpha + \beta^p \ln p_{it} + \beta^y \ln gdp_{it} + \beta^{temp} \ln temp_{it} + \beta^t t_{it} + \beta^{t^2} t_{it}^2 + v_{it} - \mu_{it} \quad (1b)$$

Correct identification of 1b is required to obtain consistent estimates of TEE. But the above equation risk suffering from bias due to a number of issues. First, national income data in Africa is of poor quality and risk being underreported. The large size nature of informal economies in Africa also increases the risk of underreporting national income data. Moreover, economic growth is not decoupled from the environment, suggesting likely mutual interdependence between income and energy consumption. This means that the income elasticity in the frontier equation is likely to suffer from bias. This study uses the *life expectancy at birth (in years)* to instrument real GDP per capita in the frontier equation. Theoretically, higher life expectancy can reduce or increase income. Lower mortality enhances the productivity of resources (most importantly human capital) and this causes income to increase. On the other hand, lower mortality may cause population growth. With fixed production factors, income will fall. Lorentzen et al. (2008), using cross-country (longitudinal) data, find that higher life expectancy leads to faster economic growth. However, Acemoglu and Johnson (2007), observe that, while improvements in life expectancy may lead to some growth in aggregate incomes, the main effect it triggers is faster population growth. Therefore, higher life expectancy may cause income per capita to decline. This suggests that the effect of life expectancy on energy consumption is more indirect through the income and population effects it generates. Consequently, life expectancy should satisfy the exclusion restriction. To test this, the authors relied on a pseudo-regression approach, because the endogenous SFA does not permit testing for the exclusion restriction. In the pseudo approach, we performed a simple regression of total energy consumption on income, price, temperature, and life expectancy and then restricted the coefficient of life expectancy at birth to zero. Failure to reject the null implies the satisfaction of the exclusion restriction. The Craig-Donald F-statistic is used to test the relevance of the instrument. For a valid instrument, the estimated F-statistics should at least be larger than Cragg-Donald's F-statistic at the 10% statistical significance level.

Second, wrong functional specification of the frontier equation can

also create bias and inconsistent results. The effect of price might depend on itself as well as on income and temperature (Henley and Peirson, 1996, 1998). Similarly, the effect of temperature on energy consumption might depend on the geographical landscape (i.e. temperate or non-temperate zones) [De Cian et al., 2007] and the level of income (De Cian et al., 2007; Petrick et al., 2010). The same observation can be deduced for income. Bearing this in mind, the log-likelihood ratio test was used to select between a Cobb-Douglas and Translog functional forms of $f(X_{it}; \beta)$. The LR statistic shows a value of 46.09 with associated p-value of 0.000, which suggests rejection of the null hypothesis in favour of the alternative hypothesis (i.e. translog functional form). The translog functional form is more flexible and helps capture complex relations, such as non-linearity, structural regime shifts and interactions between variables. Further, the variance inflation factor (VIF) was used to investigate multicollinearity in the translog frontier equation. The VIF produced an overall mean value of 1.88, which is below the threshold value of 10. Further, none of the independent variables had a VIF of >3 (see Table A1 in Appendix for results). Therefore, we estimate the translog form of eq. 1b.

Lastly, the Coelli (1995) test which has some advantages over the Schmidt and Lin (1984) skewness test was used to determine the appropriateness of the stochastic frontier function (i.e. cost or production type stochastic frontier function). According to Schmidt and Lin (1984), for a production-type SFA, the ordinary least squares residuals should be negatively skewed while for a cost-type SFA, the distribution should be positive. Checks on the ordinary least squares residuals show negative skewness (i.e. -0.4087). The estimated statistics is -5.186, which suggests rejection of the null hypothesis of no skewness against the alternative of either negative or positive skewness. These results suggest that a production-type stochastic function is appropriate.

3.2.2. Identification of inefficiency equation

The second stage of the SFA is the inefficiency equation which is modelled in this study as a function of Svirydzhenka's multidimensional measure of financial development (Eq. 1c, where γ is a constant). The financial development indicator captures the efficiency of, depth of, and access to Financial Systems, Financial Institutions, and Financial Markets (the full treatment of the derivation of these indicators is well discussed in Svirydzhenka, 2016). The use of multidimensional measure as discussed in this study is preferred to unidimensional measures of financial development. However, as discussed again in this study, relying on a comprehensive measure of financial development is not adequate to address the identification concerns originating from measurement error.

$$u_{it} = g(\gamma, FD_{it}) \quad (1c)$$

Another important source of bias which is mentioned in this study is from simultaneity bias. Thus, energy efficiency improvements can impact financial variables (see: Adom et al., 2020a; Adom, 2019; Nikolaidou and Vogiazas, 2017; Kaufmann, 2011). To address the potential bias in financial development induced by either measurement error or simultaneity bias, this study used the legal system origin of the country (i.e. whether Civil or Common law) as external instrument for financial development. According to the finance literature, significant correlation exists between financial intermediation and legal origins (La Porta et al., 1997; Beck et al., 2003, 2004; Stulz and Williamson, 2003). La Porta et al. (1997) note that differences in financial systems globally can be traced to the varying levels of protection against expropriation by insiders, as reflected by the quality of enforcement and legal rules implemented in these economies. Legal rules for protection and quality of enforcement differ systematically by legal origin, which in this study is defined as either Civil or Common law. English common laws offer more protection to shareholders and creditors but the French civil law offers the least protection (La Porta et al., 1997).

The Cragg-Donald's F-statistic was used to check for the validity of the instruments. To test for the exclusion restriction, a pseudo regression

of the estimated inefficiency in energy consumption on the instrument variable was performed, and then a test of statistical significance of the instrument was performed. These results have been reported as part of the table results. Additional identification issue relates to assumptions on TEE. In this study, we performed higher order regressions to deal with the fact that TEE may not exhibit very rapid changes at least for some countries. Further test of robustness involved using alternate measures of financial development and TEE and addressing unobserved country-specific heterogeneity bias.

3.2.3. Econometric technique: Endogenous SFA

Sections 3.2.1 and 3.2.2 present a dual identification problem, which is often complex and difficult to handle simultaneously within an SFA empirically. This might explain why previous studies have been silent on addressing endogeneity in SFA (Adom et al., 2018; Liddle and Sadorsky, 2020; Tajudeen, 2021; Sun et al., 2021, inter alia) and even in cases where some attempts are made, their approach can be described as pseudo and second-best solutions (Filippini and Zhang, 2016; Zhang and Adom, 2018). The procedures of Kutlu (2010) and Tran and Tsionas (2013) are formal procedures that treat endogeneity but only in the frontier equation. They assume that some of the regressors in the frontier equation can have correlation with the two-sided noise term but assume away any possible correlation between the two-sided error term and inefficiency term. The current study adopts the approach by Karakaplan and Kutlu (2017), which addresses the dual endogeneity problem within an SFA framework. Following this procedure, this study specifies two additional equations, (2) and (3). In eq. (2), x_{it} is the vector of endogenous variables and Z_{it} reflects $I_p \otimes z_{it}'$ vector, where z_{it}' is a vector of exogenous variables. In Eq. (3), $h_{it} = h(x_{it}, i_t' \varphi_{it}) > 0$; x_{it}, i_t' is the vector of endogenous and exogenous variables in the inefficiency equation, which excludes the constant term, and u_{it}^* is unit-specific random component assumed to be independent from v_{it} & ε_{it} .

$$x_{it} = Z_{it}\gamma + \varepsilon_{it} \quad (2)$$

$$u_{it} = h\left(x_{it}, i_t' \varphi_{it}\right) u_{it}^* \quad (3)$$

From Eqs. (1b-c) to (3), the model by Karakaplan and Kutlu suggests the following. First, the noise term (v_{it}) and inefficiency term (u_{it}) in Eq. (1b) can correlate with x_{it} but conditionally, they are independent given x_{it} and z_{it} . Second, the inefficiency term (u_{it}) and the noise term in Eq. (2) [ε_{it}] conditionally are independent given x_{it} and z_{it} (Karakaplan and Kutlu, 2017). From the above equations, Karakaplan and Kutlu (2017) estimate TEE estimate as Eq. (4), where ϕ and Φ denote the probability density function and cumulative distribution function, respectively. We used the xtsfkk code developed in Stata (Karakaplan, 2018).

$$TEE_{it} = \exp(-E[u_{it}|e_i]) = \exp\left(-h_{it}\left(\mu_{i^*} + \frac{\sigma_{i^*}\phi\left(\frac{\mu_{i^*}}{\sigma_{i^*}}\right)}{\Phi\left(\frac{\mu_{i^*}}{\sigma_{i^*}}\right)}\right)\right) \quad (4)$$

3.3. Data

This study used an unbalanced panel data of 46 African countries spanning from 1995 to 2015.¹ The choice of the data period was largely informed by the availability of consistent data for all variables and for all the countries considered in this study. Due to the data-driven nature of

¹ Angola, Cameroon, Central Africa Republic, Chad, Congo Democratic Republic, Gabon, Burundi, Comoros, Ethiopia, Kenya, Madagascar, Malawi, Mauritius, Mozambique, Rwanda, Tanzania, Uganda, Zambia, Zimbabwe, Algeria, Egypt, Libya, Morocco, Tunisia, Benin, Burkina Faso, Cape Verde, Cote d'Ivoire, Gambia, Ghana, Equatorial Guinea, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Senegal, Sierra Leone, Togo, Botswana, Lesotho, Namibia, South Africa, and Eswatini.

the SFA, obtaining sufficient and consistent data is a key requirement to force convergence. The primary variables used in this study include total primary energy consumption, total final energy consumption, the price of energy, income, temperature, financial development indicators, and legal system origin. The effect of price is measured using the real price of crude oil (i.e. international) because of the lack of consistent data on a more comprehensive measure of the energy price index for all the countries considered in this study. This is a limitation in this study, as the price effect only captures the elasticity of oil price and not total energy. At best, it captures the effect of common global shocks in the international oil market on regional energy consumption.

The effect of income is measured using data on real gross domestic product (GDP) per capita. Temperature is measured as changes in annual temperature measured in degrees Celsius. The composite index of financial system development (i.e. FD) consists of two sub-composite indices (i.e. Financial Institutions [FI] and Financial Markets [FM]). Financial institutions have evolved to include investment banks, venture capital firms, insurance companies, pension and mutual funds, and other non-bank financial institutions as major players in savings and investment decisions (Svirydzenka, 2016). The index FI consists of three other sub-indices: Financial Institution Depth (FID), Financial Institution Access (FIA), and Financial Institution Efficiency (FIE). Financial markets through stocks, bonds, wholesale money banks and by-passing traditional bank lending now permit individuals and firms to raise funds and diversify their savings (Svirydzenka, 2016). The sub-index FM consists of three other sub-indices: Financial Market Depth (FMD), Financial Market Access (FMA), and Financial Market Efficiency (FME). FID covers private sector credit to GDP, pension fund assets to GDP, mutual fund assets to GDP, and Insurance premiums, life and non-life to GDP. FIA captures bank branches per 100,000 adults and ATMs per 1000,000 adults. FIE captures net interest margin, lending-deposits spread, non-interest income to total income, overhead costs to total assets, and returns on assets and equity. FMD captures stock market capitalisation to GDP, stocks traded to GDP, international debt securities of government to GDP, and total debt securities of nonfinancial corporations to GDP. FMA captures the percent of market capitalisation outside of top 10 largest companies, and total number of issuers of debt (domestic and external, non-financial and financial corporations). FME captures stock market turnover ratio (stocks traded to capitalisation). Each of these indicators of financial development has been normalised to range between 0 and 1, where higher values suggest better performance in financial development.

Table 1 shows the descriptive statistics of all the variables used in this study. The Lm, Pesaran, and Shin approach was used to test for unit root because it does not necessarily require the panel to be strongly balanced. The demean option was used (except for the price of crude oil, which does not vary across countries), where the means of the series across panels are subtracted from the series, to mitigate the effects of cross-sectional dependence in the data. The results show that all the variables (income, price, temperature, and the indicators of financial development) are stationary in level. See Table A2 in the appendix for the test of unit root of the variables.

4. Results and discussion

This section proceeds as follows. The first part discusses the baseline results in which we assume exogeneity both in the frontier and inefficiency equations. The next sub-section presents results based on the endogenous SFA. Identification of the frontier and inefficiency equations are discussed first. This is then followed by a discussion on the frontier determinants and the estimated TEE. Next, the result of the effect of different aggregate measures of financial development (i.e. financial system development, financial institution development, and financial market development) on technical inefficiency in energy consumption is discussed. After this, there is a discussion on the effect of different aspects of financial institutions and markets on technical inefficiency in

Table 1
Descriptive Statistics of variablesVar.

Var	Unit of measure	Source	obs	Mean	Standard dev
<i>Intec</i>	Log of total primary energy consumption in Btu	The US EIA ^a	966	−2.971	1.817
<i>FD</i>	Index, range from 0 to 1	IMF ^b	924	0.141	0.096
<i>FI</i>	Index, range from 0 to 1	IMF	924	0.229	0.117
<i>FID</i>	Index, range from 0 to 1	IMF	924	0.113	0.161
<i>FIE</i>	Index, range from 0 to 1	IMF	924	0.533	0.143
<i>FIA</i>	Index, range from 0 to 1	IMF	924	0.078	0.0999
<i>FM</i>	Index, range from 0 to 1	IMF	924	0.049	0.096
<i>FMD</i>	Index, range from 0 to 1	IMF	924	0.060	0.111
<i>FME</i>	Index, range from 0 to 1	IMF	924	0.034	0.100
<i>FMA</i>	Index, range from 0 to 1	IMF	924	0.049	0.132
<i>lnpr</i>	Log of crude oil price (USD\$/b)	BP Statistics of World Review ^c	966	−0.227	0.7019
<i>lntemp</i>	Log of temperature (degrees Celsius)	The WB Climate Change Knowledge Portal	966	−0.01	0.1474
<i>lngdpc</i>	Log of GDP per capita in constant 2011 prices	World Bank	966	−0.581	1.022
<i>lnel</i>	Log of energy intensity	World Bank development indicators	966	1.5249	14.3396
<i>lnliq</i>	Log of Liquid liability % GDP	IMF ^d	966	3.1729	0.8272
<i>lnle</i>	Log of Life expectancy at birth (in years)	World Bank development indicators	966	4.025	0.1472
$legal\ system = \begin{cases} 1 & \text{if france} \\ 0 & \text{if Britain} \end{cases}$	Legal system_Britain Legal system_France	La Porta et al. (1997)	966	–	–

Source: Authors' own construction.

^a The WB Climate Change Knowledge Portal

^b Follow this url for data on financial development indicators. <https://data.imf.org/?sk=f8032e80-b36c-43b1-ac26-493c5b1cd33b>

^c <https://www.bp.com/en/global/corporate/energy-economics/statistical-review-of-world-energy.html>

^d <https://data.imf.org/>

energy consumption. The last part of this section presents some robustness checks on the results. All the table results consist of the following parts: frontier equation, technical inefficiency in energy consumption equation, regression of the variance of the two-sided error term, and diagnostic tests on the appropriateness of the instrument variables.

4.1. Baseline regression

The baseline regression assumes exogeneity both in the frontier and inefficiency equation. The frontier equation shows the level of income and its square exerts a significant positive effect on energy consumption. Price exerts negative level effect on energy consumption. The level and square of temperature exert negative effects on energy consumption. The interaction between price and temperature also exerts a statistically significant and negative effect on energy consumption. From the inefficiency equation, the mean TEE is estimated to range from 14.97% to 15.16%, suggesting inefficiency in energy consumption in the region of 84.84% to 85.03%. The effect of financial system development is negative and statistically significant. The coefficient suggests a reduction in technical inefficiency in energy consumption (TIEC) of 0.510 for every unit increase in financial system development. Development of financial institutions significantly drives TIEC downwards by 0.534 for every unit increase in the index of financial institution development. In contrast, development of financial markets drives TIEC upwards by 0.610 for every unit change in the index of financial markets development. Given that income and financial development are endogenous, the baseline coefficients of income and financial development in the frontier and inefficiency equation, respectively, and the estimate of TEE do contain some bias. We discuss the extent of this bias in the next section.

4.2. TIEC and broad-based measures of financial development

This section shows the results when broad-based measures of financial system, financial institution and financial market development are used as measures of financial development in the inefficiency equation. Table 3 shows these results. The subsequent sections discuss checks on instrument validity, frontier parameters, estimate of TEE and the effect of financial development indicators on TEE. These results are discussed with reference to the baseline regression in Table 2.

4.2.1. Identification of frontier & inefficiency equations

Consistently, the eta endogeneity test confirms the existence of endogeneity, which suggests that traditional exogenous SFA as used in Table 2 is not appropriate. For the frontier equations, life expectancy at birth meets the relevance and valid exclusion restriction criteria. These are confirmed respectively by the Cragg-Donald F-statistics (higher than the critical value at 10% statistical significance level) and the F-test in the pseudo regression involving energy consumption and life expectancy at birth (see bottom of Tables 3). Also, the legal system origin of the country satisfies the relevance criteria. The Cragg-Donald F-statistics are higher than the critical value at 10% statistical significance level. The relevance of the instrument suggests that the legal system origin of a country does correlate with the measures of financial development, which confirm the position in the finance literature. In addition, legal system origin shows no meaningful direct correlation with TIEC, albeit the valid exclusion restriction hypothesis is weakly rejected in the column of FI at 10% statistical significance level.

4.2.2. Frontier parameters

Comparing the regression output in Table 3 to the baseline regression (i.e. Table 2), the level effect of income is larger in the baseline regression by magnitudes of 0.534, 0.654 and 0.524. This shows that failing to account for endogeneity in income introduces substantial upward bias in the level income coefficient. Moreover, the square of income is positive in the baseline regression but becomes negative in Table 3. Consistently in Table 3, there is strong evidence to support an inverted U-shaped effect of income on energy consumption. This suggests the importance of structural regime shifts in explaining energy consumption in Africa. This finding supports the results of Baloch and Danish (2019), Adom (2021b), and Sbia et al. (2017). At the early stages of development, production is skewed towards energy-intensive industries, but this structure changes towards less energy-intensive industries, such as the service sector, as countries advance in economic development (Suri and Chapman, 1998; Syrquin and Chenery, 1988). The level and square terms of price and temperature are negative and but only the effect of temperature is statistically significant, suggesting that temperature does not exert a nonlinear effect on total energy consumption. However, the interactive effects of price and income and income and temperature play important roles. Both effects are negative and statistically significant (Henley and Peirson, 1996, 1998). A higher price is likely to stimulate investment in energy innovation in richer

Table 2
Baseline results: Exogenous regressors.

	(1)	(2)	(3)
	FD	FI	FM
Frontier equation			
Dep: $\ln(\text{tec})$			
Con	0.0726 (0.0997)	0.0793 (0.100)	0.0575 (0.0998)
$\ln(\text{gdp})$	1.015*** (0.0465)	0.997*** (0.0467)	1.019*** (0.0462)
$\ln(\text{pr})$	−0.0685** (0.0307)	−0.0778** (0.0308)	−0.0648** (0.0306)
$\ln(\text{temp})$	−1.370*** (0.452)	−1.454*** (0.450)	−1.344*** (0.461)
Trend	0.0214** (0.00901)	0.0230** (0.00899)	0.0217** (0.00902)
$\ln(\text{gdp})_2$	0.205*** (0.0383)	0.205*** (0.0381)	0.211*** (0.0382)
$\ln(\text{pr})_2$	−0.0513 (0.0501)	−0.0567 (0.0499)	−0.0490 (0.0500)
$\ln(\text{temp})_2$	−5.041* (3.031)	−5.300* (3.045)	−4.525 (3.005)
trend_2	0.000395 (0.000310)	0.000320 (0.000309)	0.000446 (0.000308)
$\ln(\text{gdp}) \times \ln(\text{pr})$	−0.102*** (0.0285)	−0.113*** (0.0285)	−0.0915*** (0.0280)
$\ln(\text{gdp}) \times \ln(\text{temp})$	−1.222* (0.741)	−1.090 (0.742)	−1.216 (0.745)
$\ln(\text{pr}) \times \ln(\text{temp})$	0.670*** (0.220)	0.636*** (0.219)	0.599*** (0.223)
TIEC_dep var: $\ln(\delta_{-u}^2)$			
Con	2.400*** (0.220)	2.465*** (0.221)	2.320*** (0.218)
FD	−0.510* (0.268)	−0.534*** (0.154)	
FI	—	−0.534*** (0.154)	
FM			0.604** (0.291)
Dep var: $\ln(\delta_{-v}^2)$			
Cons	−2.810*** (0.0477)	−2.820*** (0.0477)	−2.811*** (0.0477)
LL	−163.33	−159.24	−163.10
Mean_TEE	0.1516	0.1497	0.1513
Observations	924	924	924

Standard errors in parentheses. Frontier equation shows the drivers of total energy consumption. TIEC is the inefficiency equation while last part of the table shows regression of the variance in two-side error on a constant. Table 2 assumes away endogeneity in the frontier and inefficiency equations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. FD is financial system development; FI is financial Institution development, and FM is financial market development.

nations or households that have the means to invest in energy efficiency. Because richer nations or households are more likely to invest in energy efficiency, a rise in temperature might lead to a steep decline in energy consumption. This supports the findings of De Cian et al. (2007) and Petrick et al. (2010). The above results imply that income, structural changes in the economy, and temperature are key drivers of energy consumption in Africa. Consequently, growth-enhancing and climate-mitigating policies can influence energy consumption levels. However, without a correct identification strategy, these estimates, particularly income, risk suffering from bias.

4.2.3. Estimate of TEE

Figure 1 shows the estimate of TEE based on the traditional exogenous SFA and endogenous SFA. Clearly, there is a substantial upward bias in the mean estimate of TEE of about 1.58 to 3.07 percentage points. A mean difference test was performed, which produced a t-statistic value of 3.4e+03 with a p-value of 0.000. This implies that failing to account

Table 3
Endogenous regressors.

	(1)	(2)	(3)
Variables	FD	FI	FM
Frontier Equation			
Dep: $\ln(\text{tec})$			
Con	0.156 (0.213)	0.141 (0.182)	−0.0722 (0.211)
$\ln(\text{gdp})$	0.463*** (0.0976)	0.361*** (0.102)	0.495*** (0.0939)
$\ln(\text{pr})$	−0.104 (0.0660)	−0.127* (0.0703)	−0.101 (0.0656)
$\ln(\text{temp})$	−3.484*** (0.986)	−2.608*** (0.715)	−4.269*** (0.919)
Trend	0.0342* (0.0192)	0.0359* (0.0203)	0.0365* (0.0191)
$\ln(\text{gdp})_2$	−0.151** (0.0730)	−0.158** (0.0735)	−0.164** (0.0732)
$\ln(\text{pr})_2$	−0.0401 (0.109)	−0.0583 (0.116)	−0.0365 (0.108)
$\ln(\text{temp})_2$	−6.371* (3.350)	−6.774** (3.278)	−5.640* (3.326)
trend_2	0.000182 (0.000667)	2.06e-05 (0.000708)	0.000184 (0.000663)
$\ln(\text{gdp}) \times \ln(\text{pr})$	−0.240*** (0.0605)	−0.298*** (0.0638)	−0.206*** (0.0578)
$\ln(\text{gdp}) \times \ln(\text{temp})$	−3.067*** (1.166)	−1.849** (0.844)	−4.251*** (1.045)
$\ln(\text{pr}) \times \ln(\text{temp})$	0.504 (0.407)	0.225 (0.411)	0.637 (0.393)
TIEC_Dep var: $\ln(\delta_{-u}^2)$			
Con	2.588*** (0.246)	2.787*** (0.234)	2.375*** (0.233)
FD	−0.694* (0.388)		
FI		−1.294*** (0.249)	
FM			0.597* (0.329)
Dep var: $\ln(\delta_{-v}^2)$			
Cons	−2.901*** (0.0478)	−2.920*** (0.0477)	−2.898*** (0.0478)
eta1_lngdpc1	0.654*** (0.0936)	0.732*** (0.0986)	0.628*** (0.0904)
eta2_FD	−0.939 (0.598)		
eta2_FI		−1.949*** (0.454)	
Eta2_FM			0.107 (0.391)
Eta-endogeneity test	48.96***	55.75***	49.01***
Identification frontier eq			
Cragg_Donald F-stat	146.48***	177.39***	137.98***
F-test exclusion restriction	0.30	0.30	0.30
Identification Ineff eq			
Cragg_Donald F-stat	35.51***	38.55***	9.45***
F-test exclusion restriction	0.283	3.11*	0.26
LL	134.2	−10.64	−116.46
Mean_TEE	0.1190	0.1243	0.1358
Observations	924	924	924

Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Table 3 assumes endogeneity in the frontier and inefficiency equations. The legal system origin of the country is the instrument variable for financial development in the inefficiency equation. Life expectancy at birth is the instrument for real GDP per capita in the frontier Equation. The frontier equation shows the drivers of total energy consumption. TIEC is the inefficiency equation. The bottom part of the table contains information on the instrument validity test and mean estimate of TEE. FD is financial system development; FI is financial Institution development, and FM is financial market development.

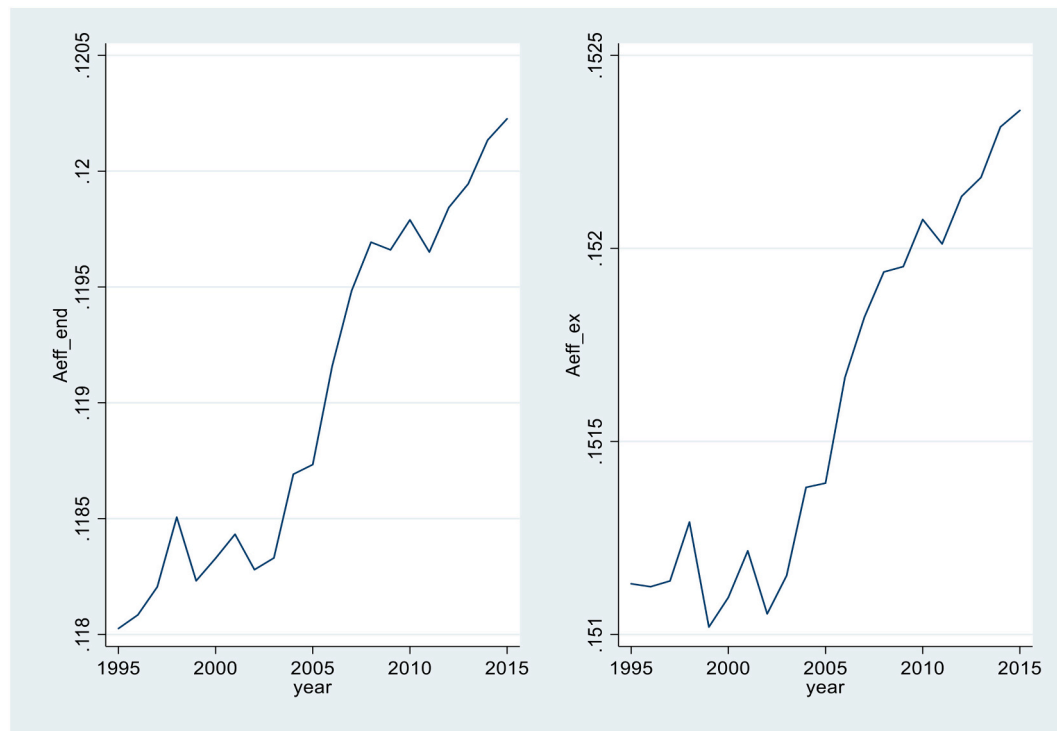


Fig. 1. Plot of annual average energy efficiency (1995–2015). Note: Aeff_end and Aeff_ex show the estimated mean technical energy consumption efficiency when we assume exogeneity and endogeneity, respectively, in both the frontier and inefficiency.

for endogeneity within the SFA introduces significant upward bias in the estimated mean TEE. Table 3 shows that the estimated mean TEE ranges from 0.119 to 0.1497 (see the bottom part of Table 3). The very low TEE indicates that a significant portion of the energy used in Africa is wasted. Based on the estimate, approximately, between 85.03% and 88.1% of the energy efficiency potential is untapped in Africa. Other studies, such as Stern (2012), Adom et al. (2018), and Agradi et al. (2022) also report very low TEE. A number of factors contribute to the extremely low TEE. They include a lack of nationwide energy efficiency policy and regulation; a lack of finance to replace obsolete equipment, structural bottlenecks, and low education and awareness on the benefits of energy efficiency. That notwithstanding, there are countries that have achieved milestones with the rollout of energy efficiency programmes. Ghana, South Africa, and Nigeria among others have rolled out programmes to replace inefficient bulbs as well as instituting efficiency labelling standards for household appliances, such as refrigerators and television.

4.2.4. TIEC and financial development Nexus

The second part of Table 3 shows the effect of financial system development (FD), financial institution development (FI), and financial market development (FM) on TIEC. We compare the results with the baseline result. In the case of using financial system development and financial institution development, the coefficients suffer substantial downward bias of 0.184 and 0.76 (in absolute terms) but for the financial market, the amount of upward bias introduced is minimal (about 0.007). According to Table 3, financial system development has a statistically significant and negative effect on TIEC. According to the estimate, an increase in financial system development by a unit is associated with a 0.694 decline in TIEC. This clearly shows that financial system development policies can help Africa deliver the SDG 7.3 target, which requires all countries to increase efforts in energy efficiency. Developing a financial system aids access to credit, particularly, when such development propels competition in the financial sector. Efficiency improvements due to competition lowers cost of capita and hence improve access to credit. Improved access to credit can boost investment

in innovation and research and development, which could help optimise the use of energy in the most efficient way. Bashir et al. (2020), Liao and Drakeford (2019), Amuakwa-Mensah et al. (2018) and Aller et al. (2018) all find that financial development reduces energy intensity, which suggests improvement in TEE.

On the effect of the development of financial institutions and the financial market on TIEC, the result shows that while the development of financial institutions exerts a negative and statistically significant effect on TIEC, the effect of the development of financial markets is rather positive and statistically significant. This means that between financial institutions and financial markets, the development of the former would rather help deliver the goal of energy efficiency. According to the coefficient value, increasing financial institution development index by one unit is associated with a 1.294 decline in TIEC. Comparatively, the growth of financial institutions in Africa has been more robust in recent years than financial markets. Several factors, such as robust economic growth, the boom in commodity prices, international trade, and urbanisation among others, have contributed to this fate. Financial institutions intermediate between savers and borrowers by diversifying risk and minimising transaction costs. Consequently, financial institutions are able to ensure a greater flow of resources, to stimulate investment in energy-efficient technologies. Cecchetti et al. (2010) note that, with a greater flow of resources, financial intermediaries are able to increase investment as well as economic growth. On the contrary, financial markets in Africa remain shallow, as they lack depth, are inefficient, less capitalised, have low liquidity, low stock market turnover ratios, limited stock listings and bond issues, and weak financial infrastructure (Abor et al., 2019). For example, Mu et al. (2013) estimate capitalisation of corporate bond in Africa to average 1.8% of GDP compared to 98.6% for the USA and 26.5% for Canada. Capitalisation of government bonds was 14.8% of GDP, which is far lower than what pertained in Central Europe and Asian countries (Abor et al., 2019). These characteristics of financial markets in Africa limit their capitalisation to finance investment activities, such as energy-efficient technologies, which are capital intensive. Similarly, Canh et al. (2020) find

that while financial institutions cause a decline in energy intensity, financial markets increase energy intensity.

4.3. TIEC and dimensions of financial institution and market

Table 4 shows the effect of different aspects of financial institution development (i.e. depth, efficiency, and access) on TIEC both for exogenous (i.e. baseline) and endogenous (i.e. final) scenarios. Consistently, the eta endogeneity test presented in the bottom part of Table 4 shows evidence in favour of an endogenous SFA. Life expectancy at birth is a valid instrument for income while legal system origin of the country is also a valid instrument for all the dimensions of financial institution development even though the claim of external validity is weak at 10% significance level for FID.

Again, the level income effect suffers from substantial upward bias while the square of income is wrongly signed in the baseline regression. This confirms the earlier claim that failing to account for endogeneity in the frontier equation can introduce bias in the income coefficient. Generally, the frontier parameters in Table 4 compare well with those obtained in Table 3, revealing robust findings. While income exerts an inverted u-shaped effect on energy consumption, the interactive effect of income and price and temperature and income are both negative and statistically significant. The estimated TEE also suffers from an upward bias of about 2 percentage points. Results based on the endogenous SFA suggest a mean inefficiency of about 87%, confirming the earlier result that significant energy efficiency potential exist in Africa.

Table 4 shows that all three aspects of financial institution development have negative and statistically significant effect on TIEC. By size, access to financial institutions exert the greatest influence on TIEC followed by the depth and efficiency of financial institutions, in that order. According to the estimate, increasing access to financial institutions is associated with a 1.227 decline in TIEC while a similar change in depth and efficiency of financial institutions is associated with 0.539 and 0.452 decline in TIEC. These coefficients when compared to the base case scenario show downward bias of 0.155 and 0.952 (in absolute terms) in FIE and FIA effects. In the case of FID, it is incorrectly signed in the base case. These results confirm the earlier claim that endogeneity of financial variables in the inefficiency equation introduces bias in the coefficient estimate. The result on FID corroborates with Adom (2021b) who find that depth measures of financial development stimulate electricity consumption efficiency. Other studies, such as Adom and Owusu-Appiah et al. (2020), Yao et al. (2021), and Chen et al. (2019a) find that financial deepening improves energy efficiency. Canh et al. (2020) find that, in the short-run, access to and depth of financial institutions reduce consumption energy intensity, but did not find evidence to support the claim that improved financial institution efficiency promote energy efficiency. On the other hand, Amuakwa-Mensah et al. (2018) find evidence supporting the energy efficiency improving effect of commercial bank efficiency. Improving access to financial institutions help remove financial constraints and thereby facilitate investment and savings. Financial institutions with higher depth are better placed to offer different financial services. Moreover, this could mean higher monetisation of the economy, which could stimulate investment and savings. Efficient provision of financial services minimises the cost of capital and facilitates technological investment, such as energy-efficient investment.

The above result implies that lowering the cost of capital (through efficiency improvement), expanding the size of the financial institution, and improving access to financial institutions can help deliver energy efficiency goals. Given that energy efficiency improvement is an important driver of energy poverty and sustainable energy for all, the result suggests that an important synergy exists between energy poverty and financial development and between sustainable energy for all and financial development. However, Africa has one of the lowest access to financial services in the world with widespread inequalities in access. Women, youth, and rural residents have very poor access compared to

their male, adult, and urban resident counterparts. Moreover, countries that are developed and have better infrastructure tend to have more access to financial services than their counterpart developing countries with infrastructural bottlenecks. At the individual country level, our results suggest that without addressing inequality in access, any policy to promote access to financial services that is not targeted would in turn create disparity in energy consumption efficiency and hence energy poverty among individuals and firms. At the regional level, the disparity in access to financial services will translate into inequality in energy consumption efficiency, making it difficult for the region to attain the Sustainable Energy for All Initiative and reduce energy poverty. Finally, the evidence of substantial bias in the coefficient estimates and estimate of TEE raises caution on previous studies' attempts to estimate TEE and the effect of financial development indicators on TEE. On the latter, it means previous studies might have weak ability to command causal inference from financial development to TEE measures.

4.4. TEE and dimensions of financial market

Table 5 shows the results for the different dimensions of financial market development. Consistently, there is evidence in favour of endogenous SFA which is indicated by the eta endogeneity test. Life expectancy at birth is a valid instrument for income (i.e. relevant and externally valid). The legal system origin of the country is externally valid but relevant only for FME and FMD. The frontier equation reveals that the level of income effect suffers from substantial upward bias with the square of income being incorrectly signed. Generally, the frontier coefficients for the endogenous SFA in Table 5 is consistent with previous results in Tables 3 and 4. Income exerts a significant inverted u-shaped effect and the interactive effects of price and income and income and temperature are negative and statistically significant. Also, the estimated TEE suffers from upward bias of about 1–2 percentage points. The estimated efficiency in energy consumption suggests inefficiency in energy consumption of about 87%. From Tables 3 to 5, we can conclude that the frontier parameters and estimated TEE are invariant to the choice of financial development indicator.

Concerning the effects of financial market dimensions on TIEC, there is no evidence to suggest that the different dimensions of financial markets significantly drive TIEC downwards. In contrast, Canh et al. (2020) find that depth of and access to financial markets significantly reduce consumption energy intensity, but efficiency of financial markets increase consumption energy intensity. The financial markets in Africa are shallow, have low depth and are inefficient, albeit, comparatively, financial markets in Africa ranks higher than the average in high-income and developing world in terms of access to financial markets (Abor et al., 2019). These characteristics of the market limits its ability to increase the flow of funds that can support technological investments. Even though compared to the baseline results, some bias can be identified, it is not a very strong evidence.

4.5. Decomposing the effects of financial development at different time-horizons

So far, it has been assumed that TEE exhibits very rapid changes. However, TEE is not likely to change rapidly at least in some economies in the short-term. Persistence in TEE data can introduce bias and weaken causal inference. We addressed this concern by first performing 3-year average generation regression. Table 6 shows the results. Again, the result shows that financial system and financial institution development rather than financial market development stimulates reductions in TIEC, but the effect of financial institution development seems statistically significant. Compared to the baseline results of Table 2, the evidence presented in Table 6 shows upward bias in the level income coefficient and the estimated coefficient of financial institution development.

Further to this, Tables 7 and 8 contain the result of a pooled regression of the estimated TEE on the initial levels of TEE and the

Table 4
TIEC and financial institution components.

	Baseline Regression			Final Regression		
	(1)	(2)	(3)	(4)	(5)	(6)
Variables	FID	FIE	FIA	FID	FIE	FIA
Frontier Equation (Dep: $\ln(\text{tec})$)						
Con	0.0473 (0.0994)	0.0586 (0.101)	0.0741 (0.0989)	0.0954 (0.172)	0.000198 (0.173)	−0.0589 (0.200)
$\ln(\text{gdp})$	1.028*** (0.0462)	1.005*** (0.0459)	1.010*** (0.0466)	0.485*** (0.0943)	0.498*** (0.0937)	0.277** (0.112)
$\ln(\text{pr})$	−0.0618** (0.0306)	−0.0828*** (0.0308)	−0.0677** (0.0306)	−0.0872 (0.0646)	−0.127** (0.0638)	−0.123 (0.0797)
$\ln(\text{temp})$	−1.332*** (0.460)	−1.600*** (0.463)	−1.329*** (0.449)	−2.650*** (0.729)	−4.136*** (0.870)	−3.214*** (0.898)
Trend	0.0219** (0.00904)	0.0247*** (0.00899)	0.0219** (0.00901)	0.0298 (0.0188)	0.0388** (0.0183)	0.0383* (0.0231)
$\ln(\text{gdp})_2$	0.206*** (0.0382)	0.204*** (0.0380)	0.203*** (0.0382)	−0.142** (0.0713)	−0.139* (0.0717)	−0.205** (0.0809)
$\ln(\text{pr})_2$	−0.0447 (0.0501)	−0.0560 (0.0497)	−0.0523 (0.0500)	−0.0416 (0.107)	−0.0491 (0.104)	−0.0499 (0.132)
$\ln(\text{temp})_2$	−4.015 (3.048)	−5.201* (3.118)	−4.616 (2.992)	−5.396 (3.291)	−7.414** (3.392)	−5.194 (3.435)
trend_2	0.000453 (0.00031)	0.000363 (0.000307)	0.000344 (0.000311)	0.000300 (0.000653)	0.000123 (0.000636)	−0.000177 (0.000810)
$\ln(\text{gdp}) \times \ln(\text{pr})$	−0.087*** (0.0282)	−0.108*** (0.0281)	−0.102*** (0.0283)	−0.225*** (0.0574)	−0.236*** (0.0573)	−0.318*** (0.0718)
$\ln(\text{gdp}) \times \ln(\text{temp})$	−1.119 (0.753)	−0.783 (0.752)	−1.178 (0.740)	−2.042** (0.835)	−3.266*** (1.033)	−3.305*** (1.003)
$\ln(\text{pr}) \times \ln(\text{temp})$	0.623*** (0.221)	0.566*** (0.220)	0.659*** (0.219)	0.261 (0.388)	0.572 (0.379)	0.471 (0.460)
TIEC_ Dep var: $\ln(\delta_{it}^2)$						
Con	2.304*** (0.218)	2.494*** (0.220)	2.361*** (0.218)	2.497*** (0.226)	2.661*** (0.247)	2.517*** (0.223)
FID	0.245 (0.173)			−0.539** (0.253)		
FIE		−0.297*** (0.0732)			−0.452*** (0.170)	
FIA			−0.275** (0.116)			−1.227*** (0.258)
Dep var: $\ln(\delta_{-v}^2)$						
Con	−2.808*** (0.0477)	−2.824*** (0.0477)	−2.813*** (0.0477)			
Dep var: $\ln(\delta_{-w}^2)$						
eta1_lngdpc1				−2.901*** (0.0477)	−2.906*** (0.0478)	−2.923*** (0.0477)
eta2_FID				0.638*** (0.0896)	0.610*** (0.0905)	0.842*** (0.110)
eta2_FIE				−1.309*** (0.378)		
Eta2_FIA					−0.467* (0.283)	
Eta-endogeneity test				52.55***	45.74***	−2.263*** (0.448)
Identification frontier eq						
Cragg_Donald F-stat				144.69***	148.06***	235.22***
F-test exclusion restriction				0.30	0.30	0.30
Identification Ineff eq						
Cragg_Donald F-stat				127.09***	8.84**	9.00**
F-test exclusion restriction				3.38*	0.64	1.65
LL	−164.15	−157.06	−162.31	−369.78	−531.18	77.02
Mean_TEE	0.1526	0.1496	0.1516	0.1331	0.1300	0.1336
Observations	924	924	924	924	924	924

Standard errors are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1. The legal system origin of the country is the instrument variable for financial development in the inefficiency equation. Life expectancy at birth is the instrument for real GDP per capita in the frontier Equation. The frontier equation shows the drivers of total energy consumption. TIEC is the inefficiency equation. The bottom part of the table contains information on the instrument validity test and mean estimate of TEE. FID is financial institution depth; FIA is financial Institution access, and FIE is financial institution efficiency.

Table 5
TIEC and components of financial markets.

	Baseline regression			Final regression		
	(1)	(2)	(3)	(4)	(5)	(6)
Variables	FMD	FME	FMA	FMD	FME	FMA
Frontier Equation: Dep- $\ln(tec)$						
<i>Con</i>	0.0490 (0.0992)	0.0614 (0.0993)	0.0667 (0.0992)	−0.131 (0.201)	−0.0432 (0.199)	0.00795 (0.227)
<i>ln(gdpc)</i>	1.014*** (0.0461)	1.025*** (0.0462)	1.031*** (0.0463)	0.503*** (0.0931)	0.500*** (0.0937)	0.503*** (0.102)
<i>ln(pr)</i>	−0.0688** (0.0305)	−0.0618** (0.0306)	−0.0578* (0.0306)	−0.106 (0.0650)	−0.0981 (0.0657)	−0.0915 (0.0653)
<i>ln(temp)</i>	−1.391*** (0.463)	−1.324*** (0.458)	−1.259*** (0.453)	−4.513*** (0.961)	−4.250*** (0.905)	−3.903*** (0.961)
<i>Trend</i>	0.0219** (0.00898)	0.0208** (0.00902)	0.0195** (0.00903)	0.0372** (0.0189)	0.0355* (0.0191)	0.0330* (0.0190)
<i>ln(gdpc)_2</i>	0.220*** (0.0382)	0.207*** (0.0383)	0.211*** (0.0383)	−0.148** (0.0731)	−0.165** (0.0731)	−0.155** (0.0729)
<i>ln(pr)_2</i>	−0.0476 (0.0498)	−0.0468 (0.0502)	−0.0461 (0.0501)	−0.0351 (0.107)	−0.0346 (0.108)	−0.0335 (0.108)
<i>ln(temp)_2</i>	−4.236 (2.995)	−4.631 (3.012)	−4.522 (2.996)	−5.630* (3.309)	−5.858* (3.341)	−5.934* (3.336)
<i>trend_2</i>	0.000482 (0.000307)	0.000453 (0.000309)	0.000496 (0.000309)	0.000212 (0.000656)	0.000198 (0.000664)	0.000278 (0.000663)
<i>ln(gdpc) x ln(pr)</i>	−0.0916*** (0.0279)	−0.0910*** (0.0280)	−0.0878*** (0.0280)	−0.201*** (0.0570)	−0.209*** (0.0576)	−0.208*** (0.0591)
<i>ln(gdpc) x ln(temp)</i>	−1.046 (0.751)	−1.279* (0.744)	−1.238* (0.739)	−4.351*** (1.124)	−4.221*** (0.999)	−3.903*** (1.032)
<i>ln(pr) x ln(temp)</i>	0.531** (0.224)	0.653*** (0.221)	0.682*** (0.221)	0.616 (0.390)	0.681* (0.390)	0.661* (0.393)
TIEC_ Dep var: $\ln(\delta_{-u}^2)$						
<i>Con</i>	2.321*** (0.218)	2.330*** (0.217)	2.359*** (0.218)	2.347*** (0.229)	2.398*** (0.229)	2.440*** (0.251)
<i>FMD</i>	0.627*** (0.182)			0.654** (0.280)		
<i>FME</i>		−0.00252 (0.126)			0.0489 (0.143)	
<i>FMA</i>			−0.799* (0.417)			−0.696 (0.735)
Dep var: $\ln(\delta_{-w}^2)$						
				−2.902*** (0.0478)	−2.893*** (0.0478)	−2.896*** (0.0479)
Dep var: $\ln(\delta_{-v}^2)$						
<i>Con</i>	−2.820*** (0.0477)	−2.805*** (0.0477)	−2.809*** (0.0477)			
<i>eta1_lngdpc1</i>				0.616*** (0.0899)	0.630*** (0.0903)	0.627*** (0.0989)
<i>eta2_FMD</i>				0.276 (0.365)		
<i>eta2_FME</i>					−0.00104 (0.220)	
<i>Eta2_FMA</i>						−0.134 (0.647)
Eta-endogeneity test				46.98***	48.86***	47.81***
Identification frontier eq						
Cragg_Donald F-stat				138.35***	141.05***	144.58***
F-test exclusion restriction				0.30	0.30	0.30
Identification Ineff eq						
Cragg_Donald F-stat				7.45**	10.76***	2.08
F-test exclusion restriction				0.37	0.32	0.64
LL	−159.42	−165.15	−163.15	−250.96	−238.88	−512.35
Mean_TEE	0.1501	0.1524	0.1529	0.1398	0.1347	0.1322
Observations	924	924	924	924	924	924

Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The legal system origin of the country is the instrument variable for financial development in the inefficiency equation. Life expectancy at birth is the instrument for real GDP per capita in the frontier Equation. The frontier equation shows the drivers of total energy consumption. TIEC is the inefficiency equation. The bottom part of the table contains information on the instrument validity test and mean estimate of TEE. FMD is financial market depth; FMA is financial market access, and FME is financial market efficiency.

Table 6
three-year average regression.

Variables	(1) FD	(2) FI	(3) FM
Frontier Equation: Dep- $\ln(tec)$			
<i>Con</i>	−0.000961 (0.326)	0.00203 (0.304)	−0.244 (0.367)
<i>ln(gdpc)</i>	0.575*** (0.158)	0.505*** (0.157)	0.545*** (0.164)
<i>ln(pr)</i>	−0.130 (0.171)	−0.162 (0.174)	−0.117 (0.184)
<i>ln(temp)</i>	−5.067*** (1.475)	−4.822*** (1.259)	−5.608*** (1.473)
<i>Trend</i>	0.0327 (0.0367)	0.0350 (0.0374)	0.0357 (0.0399)
<i>ln(gdpc)_2</i>	−0.216* (0.130)	−0.199 (0.122)	−0.267** (0.135)
<i>ln(pr)_2</i>	−0.0952 (0.329)	−0.114 (0.334)	−0.0828 (0.356)
<i>ln(temp)_2</i>	−20.67** (8.095)	−23.01*** (6.823)	−15.04** (7.639)
<i>trend_2</i>	0.000232 (0.00136)	9.78e-05 (0.00139)	0.000174 (0.00148)
<i>ln(gdpc) x ln(pr)</i>	−0.238** (0.110)	−0.297*** (0.113)	−0.209* (0.115)
<i>ln(gdpc) x ln(temp)</i>	−4.777** (2.255)	−3.241* (1.794)	−6.462*** (2.045)
<i>ln(pr) x ln(temp)</i>	1.377* (0.750)	1.084 (0.755)	1.401* (0.792)
TIEC_Dep var: $\ln(\delta_{-w}^{-2})$			
<i>Con</i>	2.319*** (0.263)	2.499*** (0.251)	2.127*** (0.249)
<i>FD</i>	−0.520 (0.675)		
<i>FI</i>		−1.069** (0.429)	
<i>FM</i>			1.078 (0.695)
Dep var: $\ln(\delta_{-w}^{-2})$			
	−3.153*** (0.0880)	−3.172*** (0.0874)	−3.148*** (0.0886)
<i>eta1_lngdpc1</i>	0.661*** (0.159)	0.708*** (0.154)	0.703*** (0.162)
<i>eta2_FD</i>	−0.211 (0.882)		
<i>eta2_FI</i>		−1.374* (0.740)	
<i>Eta2_FM</i>			0.739 (0.677)
Eta-endogeneity test	17.37***	21.25***	47.81***
Identification_frontier eq			
Cragg_Donald F-stat	45.32***	50.05***	42.70***
F-test exclusion restriction	1.15	1.15	1.15
Identification_Ineff eq			
Cragg_Donald F-stat	11.53**	13.12***	3.20
F-test exclusion restriction	0.45	1.35	0.02
LL	−29.92	−34.76	−512.35
Mean_TEE	0.1648	0.1669	0.1322
Observations	308	308	308

Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The legal system origin of the country is the instrument variable for financial development in the inefficiency equation. Life expectancy at birth is the instrument for real GDP per capita in the frontier Equation. The frontier equation shows the drivers of total energy consumption. TIEC is the inefficiency equation. The bottom part of the table contains information on the instrument validity test and mean estimate of TEE. FD is financial system development; FI is financial Institution development, and FM is financial market development.

indicator of financial development for the periods 2015/1995, 1999/1995, 2004/2000, 2009/2005, and 2015/2010. Here, we observe whether successive changes in TEE are associated with changes in financial development. Because changes in TEE are used as the dependent variable, evidence of causal effect implies that positive changes in TEE should be associated with positive changes in financial development. The result shows that a 20-year average change in TEE (2015/1995) is largely associated with a change in financial system and institution development. This is shown by the positive effect of financial system and institution development on 20-year average change in TEE. The time horizon was further shortened to ascertain the stability of the effect of financial system and institution development on changes in TEE. Except for the period, 2000/2004, where positive changes in financial system and institution development are associated with negative changes in TEE, generally, positive changes in financial system and institution development are followed by positive changes in TEE. These results suggest strong causal inferences from changes in financial development to TEE improvements.

4.6. Robustness checks

Several issues can compromise the validity of the results obtained in this study. Therefore, in this section, we subject the results to several of these issues to ascertain the robustness of the results obtained.

4.6.1. Using specific outcome measure of financial development

A number of robustness checks were performed to substantiate the claim of causal inference from financial development to TEE. First, Svirydzienka's multidimensional measure of financial system development has some limitations that could have implications for the results. Despite being broad, the FD measure still ignores important developments in the financial sector, such as credit transfers, direct-debits, and mobile money banking. It also ignores diversity in the types of financial intermediation and complexity in institutions and investment. The index only captures characteristics of financial systems and not outcomes of financial systems. Moreover, the FD index may overstate the true extent of financial development because, for some countries, government controls can influence the efficiency of financial systems. Lastly, higher values of the index may not necessarily suggest better performance. According to Svirydzienka (2016), it may suggest that the financial system of a country has been overstretched beyond its structural and regulation capabilities, and this could have a negative effect on growth and stability. These problems imply that measurement error might still be present in the index. Using the legal system origin as an instrument for financial development is one way we have demonstrated in this study can address this concern. Further, this study resorted to using a standard measure of financial development (i.e. liquid liability as a percent of GDP). Liquid liability covers the sum of currency and deposits in the central bank (M0), plus transferable deposits and electronic currency (M1), plus time and savings deposits, foreign currency transferable deposits, certificates of deposit, and securities repurchase agreements (M2), plus travelers checks, foreign currency time deposits, commercial paper and shares of mutual funds or market funds held by residents. Higher values suggest higher monetization of the economy, and this is comparable across all countries. Table 9 shows the results. The result remains robust. High monetization of economies significantly reduces TIEC. This underscores the fact that measurement errors in FD do not drive our result.

4.6.2. Addressing short- and long-run inefficiency and using energy intensity as an alternative measure

Second, our estimate of TEE is likely to suffer from bias because it does not separate transient and persistent inefficiency as well as unobserved heterogeneity. This study applied the Kumbhakar et al. (2014) approach, which separates unobserved heterogeneities from transient and persistent inefficiency to estimate TEE. The correlation coefficient

Table 7

FD effect over different time horizon.

Dep var.: TEE_DF					
	(1)	(2)	(3)	(4)	(5)
Variables	2015/1995	1999/1995	2000/2004	2009/2005	2015/2010
TEE_1995	0.00603*** (0.00160)	0.00180*** (0.000612)			
TEE_2000			0.00187*** (0.000243)		
TEE_2005				0.00102 (0.000633)	
TEE_2010					0.00122*** (0.000300)
FD	0.0294*** (0.00322)	0.0168*** (0.00173)	−0.00287*** (0.000381)	0.0142*** (0.00108)	−0.000102 (0.000383)
Constant	−0.00267*** (0.000390)	−0.00238*** (0.000212)	0.000548*** (5.66e-05)	−0.00120*** (0.000123)	0.000305*** (6.05e-05)
Observations	924	924	924	924	924
R-squared	0.457	0.410	0.085	0.452	0.045

Robust standard errors in parentheses: TEE_DF measures the change in TEE between the ending year and the initial year. TEE-1995 – TEE-2010 gives the initial TEE for those years. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8

FI effect over different time horizon.

Dep var.: TEE_DF					
	(1)	(2)	(3)	(4)	(5)
Variables	2015/1995	1999/1995	2000/2004	2009/2005	2015/2010
TEE_1995	0.0106*** (0.00164)	0.00451*** (0.000661)			
TEE_2000			0.00143*** (0.000212)		
TEE_2005				0.00320*** (0.000601)	
TEE_2010					0.00117*** (0.000294)
FI	0.0194*** (0.00239)	0.0103*** (0.00131)	−0.00198*** (0.000265)	0.00970*** (0.000723)	0.000220 (0.000245)
Constant	−0.00348*** (0.000523)	−0.00269*** (0.000296)	0.000646*** (6.87e-05)	−0.00167*** (0.000155)	0.000246*** (6.87e-05)
Observations	924	924	924	924	924
R-squared	0.401	0.323	0.078	0.401	0.045

Robust standard errors in parentheses: TEE_DF measures the change in TEE between the ending year and the initial year. TEE-1995 – TEE-2010 gives the initial TEE for those years. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

between our initial estimate of TEE and persistent TEE is 71% approximately, but the correlation with transient TEE is very weak, 3%. Thus, our estimated TEE reflect more persistent TEE than transient TEE. This confirms our claim that the estimated TEE do not exhibit very rapid changes. However, TEE based on Kumbhakar et al. approach does not address endogeneity, and therefore, suffer from bias. To deal with measurement issues in TEE, this study used log of energy intensity as the dependent variable. In this regression, we controlled for year effect to deal with time-varying common shocks in the data, which might impair causal inference from financial development to TEE. Table 10 contains the result.² All the indicators of financial development have negative and statistically significant effects (except for FM case) on the log of energy intensity. This supports the earlier result that between financial market and financial institutions, it is the development of the latter that drives energy efficiency. Further, the result confirms the earlier

conclusion that all aspects of financial institution development but most importantly access to financial institutions drive energy efficiency upward. The above is an indication that errors associated with the measure of TEE do not drive the result.

4.6.3. Accounting for the level of economic development

Third, context heterogeneity might drive the result. Energy efficiency improvements require a switchover to new improved technologies, and context heterogeneities might either propel or inhibit this process. Possessing worthy collateral is an important precondition to access capital from financial systems. However, poor nations or individuals compared to their rich counterparts might not possess these worthy collaterals. Thus, whether financial development could stimulate investment in energy efficiency might depend on the level of development of the country. Failing to account for this omission could have biased our results. To address this concern, we grouped the sample

² In Tables 11 to 12, we only report results on inefficiency equation just to keep to the details.

Table 9
Specific measure of Financial Development and TIEC.

Variables	Liquid Liability
Dependent variable	ln(tec)
<i>Frontier Equation</i>	
Con	−0.042 (0.171)
ln(gdpc)	0.549*** (0.109)
ln(pr)	−0.097 (0.061)
ln(temp)	−4.291*** (0.868)
Trend	0.035** (0.018)
ln(gdpc)_2	−0.163** (0.078)
ln(pr)_2	−0.036 (0.100)
ln(temp)_2	−7.131** (3.318)
trend_2	0.000 (0.001)
ln(gdpc) × ln(pr)	−0.214*** (0.057)
ln(gdpc) × ln(temp)	−4.042*** (0.927)
ln(pr) × ln(temp)	0.727** (0.366)
TIEC_Dep var:ln(δ_{-u}^2)	
Con	2.548*** (0.248)
ln(lliq)	−0.053* (0.032)
Dep var:ln(δ_{-u}^2)	
Con	−2.919*** (0.047)
eta2(lngdpc)	0.594*** (0.103)
eta2(lliq)	−0.007 (0.042)
Eta-endogeneity test	45.02***
Identification of frontier_eq	
Cragg-Donald F-test	451.40***
F-test Exclusion restriction	0.30
Identification of inefficiency_eq	
Cragg-Donald F-test	20.73***
F-test Exclusion restriction	0.33
Observations	966
log likelihood	−2173.73
Mean ECE	0.1420

Standard errors are in parentheses. Symbols indicate significance at the 1% (***), 5% (**), and 10% (*) levels. Legal system origin is the instrument variable for financial development in inefficiency equation. Life expectancy at birth is the instrument for real GDP per capita in the frontier Equation.

into Low-income countries (LICs) and Middle (lower and upper) – and High-income countries (MHICs).³ Table 11 shows the results for broad-based measures of financial development while Table 12 shows the results for the dimensions of financial institution development. Tables 11 and 12 reveal that, both in LICs and MHICs, financial system and institution development rather than financial market development reduce TIEC and promote TEE. Concerning the aspects of financial institution

development, the result shows that all aspects of financial institutions but most importantly access to finance drive TIEC downwards. These results indicate that context heterogeneity bias does not drive the result.

Comparing LICs to MHICs, we find that financial development in LICs might be more rewarding in terms of promoting TEE. This assertion is true whether we consider broad-based measures of financial development or specific components of financial institutional development. LICs markets are flooded with inefficient second-hand technologies. Moreover, structural bottlenecks such as lack of proper institutions make it relatively difficult to switch from inefficient obsolete equipment and appliances to newly improved technologies. Preliminary test shows that MHICs perform better on all the indicators of financial development than LICs do (see Table A3 in Appendix). These conditions imply that the marginal productivity of introducing new technology in LICs is higher than MHICs. Development of the financial sector help remove the financial constraint barrier and thereby help transition process from low-efficient appliance stock to high-efficient appliance stock. During this transition period, where individuals and firms virtually replace old inefficient appliance stock, energy efficiency would experience a significant boost, all things being equal, in LICs that suffer from lack of finance. However, further development of financial systems might not exert the same effect on the efficiency level of the appliance stock, rather it might increase the appliance stock (i.e. causing scale effects). This could even be cascaded by possible rebound effects, as savings accrued from energy savings could, among other things, increase the purchase of energy-using appliances. Consequently, the transition from low-efficient appliance stock to high-efficient appliance stock induced by financial development is likely to boost TEE significantly more in LICs than MHICs. Our result points to the important fact that deepening financial development especially in LICs could significantly spur convergence in energy efficiency goals, which is vital to the Sustainable Energy for All agenda. In this regard, the result presented in Table 12 shows that while improving the depth of and access to financial institutions might prove more relevant for LICs, for MHICs, it is the improvement in the efficiency of and access to financial institutions that matter. These results imply that policies on financial development targeted at improving energy outcomes should be mindful of the context heterogeneity.

5. Conclusion and policy implications

The existing literature on the nexus between energy efficiency and financial development is nascent but the result remains inconclusive. A number of factors such as identification problem, misspecification, measurement problems and heterogeneity bias underpin the result. This study explored the extent of bias originating from endogeneity (due to measurement error and simultaneity bias) and heterogeneity bias. A number of strategies were adopted to improve the causal inference from financial development to energy consumption efficiency. First, external instruments (i.e. legal system origin and life expectancy at birth) were used to instrument endogenous financial development in the inefficiency equation and endogenous income in the frontier equation, respectively. We have shown that, largely, both instruments satisfy the conditions of relevance and validly excludable instrument. Additional strategies adopted in this study included performing regression at much higher levels to address the concern that inefficiency in energy consumption may not exhibit very rapid changes at least for some other countries in the short-term. Further robustness checks were performed to substantiate the reliability of the result obtained. Our estimation approach is based on the endogenous stochastic frontier technique using data for 46 African countries that spanned from 1995 to 2015. The following results emerged from the study.

First, we have shown that failing to address identification concerns introduce substantial bias in the estimate of TEE, income elasticity in the frontier equation, and the effect of financial development on TEE. Largely, the estimate of TEE and income elasticity suffers from an upward bias while the effect of financial development suffers from a

³ The pooling of middle and high-income countries together is due to data limitations.

Table 10
Using energy intensity (Inei) as dependent variable.

	(1)	(2)	(3)	(4)	(5)	(6)
Variables	EI	EI	EI	EI	EI	EI
<i>FD</i>	−13.86*** (2.603)					
<i>FI</i>		−15.59*** (3.345)				
<i>FM</i>			−4.138 (3.019)			
<i>FID</i>				−7.540*** (1.322)		
<i>FIE</i>					−7.661*** (1.763)	
<i>FIA</i>						−24.19*** (8.078)
<i>Constant</i>	4.285*** (0.715)	6.123*** (1.008)	2.299*** (0.673)	3.108*** (0.652)	6.168*** (1.105)	5.261*** (1.167)
Year Effect	YES	YES	YES	YES	YES	YES
Observations	924	924	924	924	924	924
R-squared	0.032	0.039	0.024	0.030	0.029	0.049

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11
Results – by level of economic development.

	High-Middle Income Countries			Low-Income Countries		
	(1)	(2)	(3)	(4)	(5)	(6)
Variables	FD	FI	FM	FD	FI	FM
TIEC_Dep var:ln(δ_u^{-2})						
<i>Con</i>	2.509*** (0.302)	2.499*** (0.251)	2.329*** (0.284)	2.329*** (0.401)	2.051*** (0.392)	2.380*** (0.408)
<i>FD</i>	−0.632 (0.473)			−1.947** (0.859)		
<i>FI</i>		−1.069** (0.429)			−1.244** (0.597)	
<i>FM</i>			0.499 (0.394)			0.179 (1.085)
Dep var:ln(δ_u^{-2})						
	−2.777*** (0.0589)	−3.172*** (0.0874)	−2.772*** (0.0589)	−3.531*** (0.0818)	−3.510*** (0.0818)	−3.533*** (0.0821)
<i>eta1_lngdpc1</i>	0.435*** (0.0666)	0.456*** (0.0758)	0.436*** (0.0721)	2.714*** (0.271)	3.665*** (0.375)	1.409* (0.853)
<i>eta2_FD</i>	−0.197 (0.679)			−3.757*** (1.193)		
<i>eta2_FI</i>		−0.743 (0.541)			−1.946** (0.773)	
<i>Eta2_FM</i>			0.326 (0.440)			−3.136*** (1.149)
Eta-endogeneity test	44.01***	37.82***	37.62***	116.34***	112.44***	15.87***
Identification_frontier eq						
Cragg_Donald F-stat	45.32***	124.31***	122.48***	36.63***	59.74***	41.72***
F-stat_test exclusion restriction	0.34	0.34	0.34	0.40	0.40	0.40
Identification_Ineff eq						
Cragg_Donald F-stat	14.32***	31.57***	10.36***	26.35***	12.26***	35.52***
F-stat_test exclusion restriction	0.30	1.11	0.85	4.18	0.11	11.94***
LL	1.61	−94.73	−161.17	434.13	421.72	352.32
Mean_TEE	0.1590	0.1568	0.1684	0.1710	0.2316	0.1381
Observations	609	609	609	315	315	315

Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Legal system origin of the country is the instrument variable for financial development in the inefficiency equation. Life expectancy at birth is the instrument for real GDP per capita in the frontier Equation. **This table provides results when we use the broad-based measures of financial development.**

significant downward bias. These results raise caution about future studies' estimates of the effect of financial development on TEE. We have demonstrated several ways to address the potential bias in future related studies. However, particularly, concerning the use of external instruments, it might be considered as luxury in some instances. Thus,

there is no one-size-fits-all approach, as context characteristics, data availability, and theory might play a significant role. Consequently, we do not claim that using the legal system origin of countries to instrument financial development might work for all cases. It should be tested in each case. Moreover, performing regressions at much higher levels was

Table 12
TIEC and Dimensions of FI.

Variables	High-Middle Income Countries			Low-Income Income Countries		
	(1)	(2)	(3)	(4)	(5)	(6)
	FID	FIE	FIA	FID	FIE	FIA
TIEC_Dep var:ln(δ_w^{-2})						
<i>Con</i>	2.434*** (0.283)	2.694*** (0.303)	2.486*** (0.273)	1.981*** (0.385)	1.897*** (0.418)	2.106*** (0.385)
<i>FID</i>	−0.218 (0.428)			−1.865*** (0.690)		
<i>FIE</i>		−0.557*** (0.209)			−0.103 (0.173)	
<i>FIA</i>			−0.783*** (0.256)			−2.955*** (0.804)
Dep var:ln(δ_w^{-2})						
	−2.772*** (0.0588)	−2.796*** (0.0588)	−2.789*** (0.0588)	−3.512*** (0.0819)	−3.497*** (0.0817)	−3.540*** (0.0820)
<i>eta1_lngdpc1</i>	0.430*** (0.0733)	0.378*** (0.0725)	0.553*** (0.0863)	3.459*** (0.970)	4.247** (1.914)	2.716*** (0.979)
<i>eta2_FID</i>	−0.383 (0.757)			−2.458*** (0.768)		
<i>eta2_FIE</i>		−0.290 (0.331)			−0.345 (0.241)	
<i>Eta2_FIA</i>			−1.475*** (0.496)			−1.871 (1.238)
Eta-endogeneity test	38.09***	27.20***	41.22***	21.96***	7.05**	11.68***
Identification frontier eq						
Cragg_Donald F-stat	150.81***	122.70***	153.52***	57.69***	18.15***	9.12**
F-stat_test exclusion restriction	0.34	0.34	0.34	0.40	0.40	0.40
Identification Ineff eq						
Cragg_Donald F-stat	96.47***	12.33***	9.73***	27.35***	22.47***	82.91***
F-stat_test exclusion restriction	0.50	0.05	1.10	0.02	3.58*	2.54
LL	−309.03	−369.68	−88.76	323.02	97.53	678.93
Mean_TEE	0.1626	0.1635	0.1621	0.2310	0.2553	0.2002
Observations	609	609	609	135	135	135

Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Legal system origin of the country is the instrument variable for financial development in the inefficiency equation. Life expectancy at birth is the instrument for real GDP per capita in the frontier Equation.

premised on the assumption that TEE does not exhibit very rapid changes at least for some countries in the short-term. While we do not expect energy efficiency to exhibit very rapid changes in the short-term, there might be an exception to the rule, particularly in cases where markets and institutions are not properly developed. Therefore, adopting this approach should be case informed.

Regarding the influence of financial development on TEE, we have revealed that, broadly, financial system development (i.e. constellation of financial institutions and markets) drives TEE positively. However, between financial institutions and markets, it is the development of the former and not the latter that drives TEE positively. By implication, whether one observes an effect or not on energy consumption efficiency depends on the measure of financial development. For the different sub-indices of financial institutions, this study has shown that access to, efficiency of, and depth of financial institutions all drive TEE upward, but the effect of access is more. This implies that financial institution policies targeted at improving access, efficiency, and depth of financial institutions are important for achieving the goals of energy efficiency and by extension energy poverty and SDG 7. However, existing inequalities in energy poverty and financial inclusion levels among countries and individuals in Africa, imply that implementing an untargeted and uncoordinated financial policy can deepen the inequality already prevalent in energy poverty. Because energy indicators have important links with socio-economic factors, financial policies on energy-related issues that fail to take cognisance of these existing

inequalities can further widen social exclusion. Therefore, the implementation of financial institution policy should be well coordinated and targeted. Financial institution development policies targeting socially excluded groups such as women, youth, and rural folks at the individual country levels and poorer nations at the regional level could significantly narrow the inequality gap already existing in energy efficiency and energy poverty and hence drive the Sustainable Energy for All agenda. As revealed in this study, energy efficiency gains associated with developing the financial sector are likely to be heightened among low-income countries. Lastly, policymakers should be mindful of the heterogeneities that exist among countries, groups, and individuals when designing financial policies targeted at improving energy outcomes.

This study considered aggregate energy consumption efficiency and how financial development influences it. An important future extension could examine the same issue first at a more micro level and second for different energy types. This is because technological requirements as well as the accompanying investment size might be different.

CRediT authorship contribution statement

Philip Kofi Adom: Conceptualization, Methodology, Data curation, Writing – original draft, Writing – review & editing. **Franklin Amuakwa-Mensah:** Conceptualization, Writing – original draft, Writing – review & editing. **Charity Dzifa Akorli:** Conceptualization,

Writing – original draft, Writing – review & editing.

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Appendix A. Appendix

Table A1. VIF test of multicollinearity.

Variable	VIF	1/VIF
$\ln(gdpc)$	1.57	0.635
$\ln(pr)$	2.38	0.421
$\ln(temp)$	2.81	0.356
$0.5 * \ln(gdpc) * \ln(pr)$	1.86	0.538
$0.5 * \ln(gdpc) * \ln(temp)$	1.70	0.587
$0.5 * \ln(pr) * \ln(temp)$	1.25	0.803
$0.5 * (\ln(gdpc))^2$	1.43	0.698
$0.5 * (\ln(pr))^2$	2.03	0.493
$0.5 * (\ln(temp))^2$	1.93	0.519

This table presents the result on the VIF test of multicollinearity. The VIF for all the regressors in the frontier equation is less than the threshold value of 10, which suggest no multicollinearity.

Table A2: Test of Unit root robust to cross sectional dependence.

Variables	Statistic	P-value
$\ln(tec)$	−4.8847	0.000
$\ln(pr)$	−3.8634	0.0001
$\ln(gdpc)$	−2.6039	0.0046
$\ln(temp)$	−14.9853	0.000
$\ln(tfec)$	−1.561	0.0590
$\ln(liq)$	−3.546	0.000
$\ln(ei)$	−11.396	0.000
FD	−8.1400	0.000
FI	−8.9498	0.000
FIA	−1.4705	0.0707
FID	−5.7435	0.000
FIE	−9.8938	0.000
FM	−2.8867	0.0019
FMA	−5.4090	0.000
FMD	−3.9270	0.000
FME	−4.7127	0.000

Null hypothesis: All panels contain unit root.

Alternative hypothesis: some panels are stationary.

The Lm, Pesaran and Shin unit root test suggests rejection of the null of unit root in favour of the alternative of no unit root.

Table A3. Mean difference test of level of financial development.

	LICs	MHICs	Combined	Difference
FD	0.1078 (0.0023)	0.1587 (0.0043)	0.1414 (0.0032)	−0.0509*** (0.0065)
FI	0.1767 (0.0026)	0.2553 (0.0054)	0.2285 (0.0039)	−0.0786*** (0.0077)
FM	0.0349 (0.0032)	0.0561 (0.0044)	0.0489 (0.0031)	−0.0212** (0.0064)
FID	0.06414 (0.0031)	0.1387 (0.0077)	0.1133 (0.0053)	−0.0746*** (0.0109)
FIA	0.0435 (0.0021)	0.0953 (0.0047)	0.0777 (0.0033)	−0.0518*** (0.0067)
FIE	0.4678 (0.0071)	0.5663 (0.0056)	0.5327 (0.0047)	−0.0984*** (0.0094)
FMD	0.0530 (0.0044)	0.0642 (0.0050)	0.0604 (0.0036)	−0.0113 (0.0077)
FMA	0.0342 (0.0069)	0.0572 (0.0055)	0.0494 (0.0043)	−0.0230* (0.0091)
FME	0.0135 (0.0022)	0.0443 (0.0048)	0.0338 (0.0033)	−0.0308*** (0.0069)
Observation	315	609	924	

Standard errors are in parentheses. Symbols indicate significance at the 0.1% (***), 1% (**), 5% (*), and 10% (†) levels.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2023.106577>.

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